

BIO262 Physiology

Module 2

Cells and their parts

Osmosis and diffusion

How cells make proteins

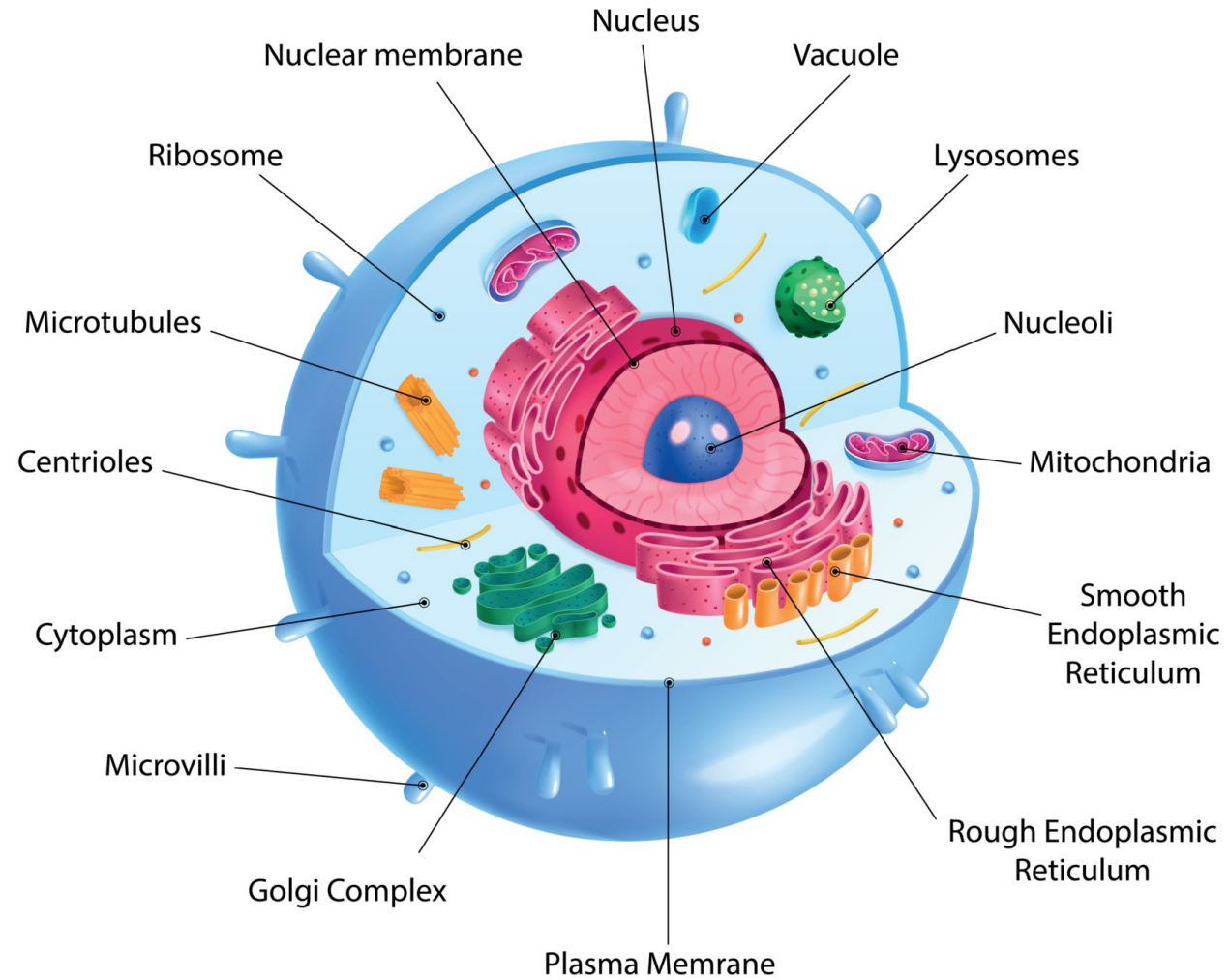
HUMAN CELL ANATOMY

Three MAIN parts:

- Plasma/cell membrane (phospholipid bilayer)
- Cytoplasm (full of organelles)
- Nucleus (full of DNA)

Humans have two main types:

- Somatic (body)
- Sex/gametes (reproductive)

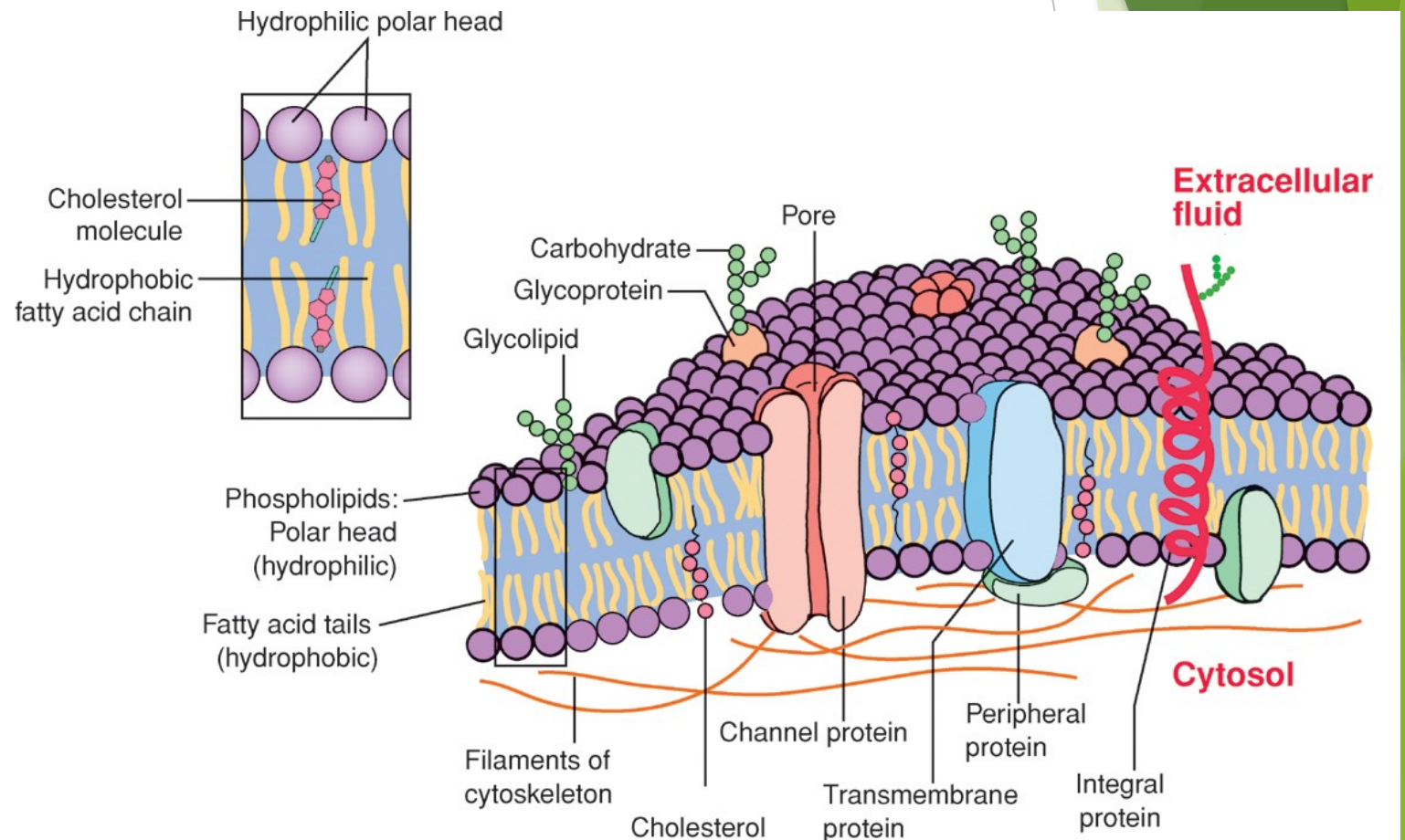


Plasma Membrane Structure & Function

Main Function is to separate extracellular environment from intracellular. Eg: pH – outside 7.35-7.45, inside 6.8

Semipermeable membrane!
Selective!!! DUE to LIPID tails (phospholipids with cholesterol) – Non polar and lipid soluble molecules can enter/exit without any 'help'. Eg: O₂, CO₂, fats, some hormones.....

Polar molecules, ions, water soluble molecules need 'help'.

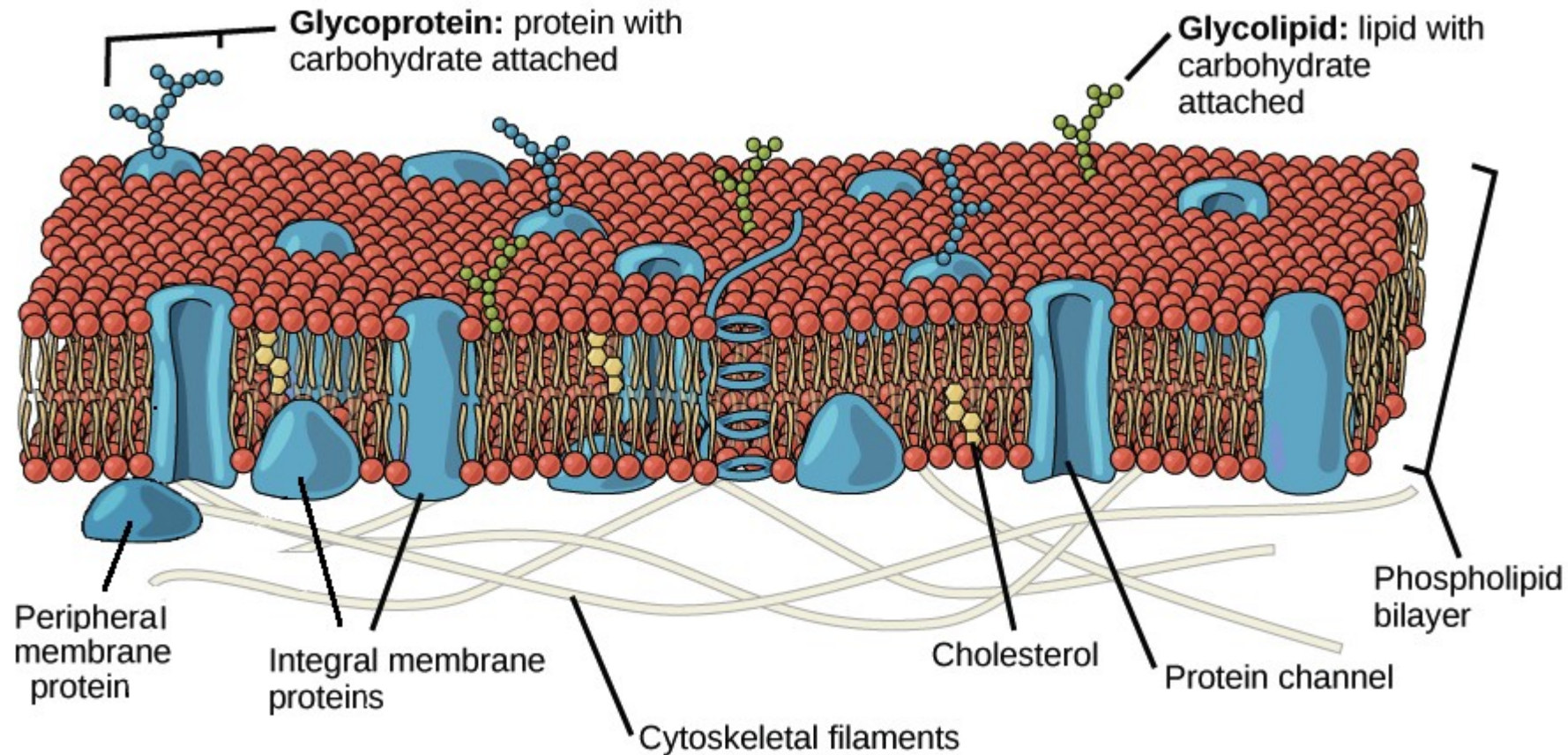


Plasma Membrane Proteins and their functions:

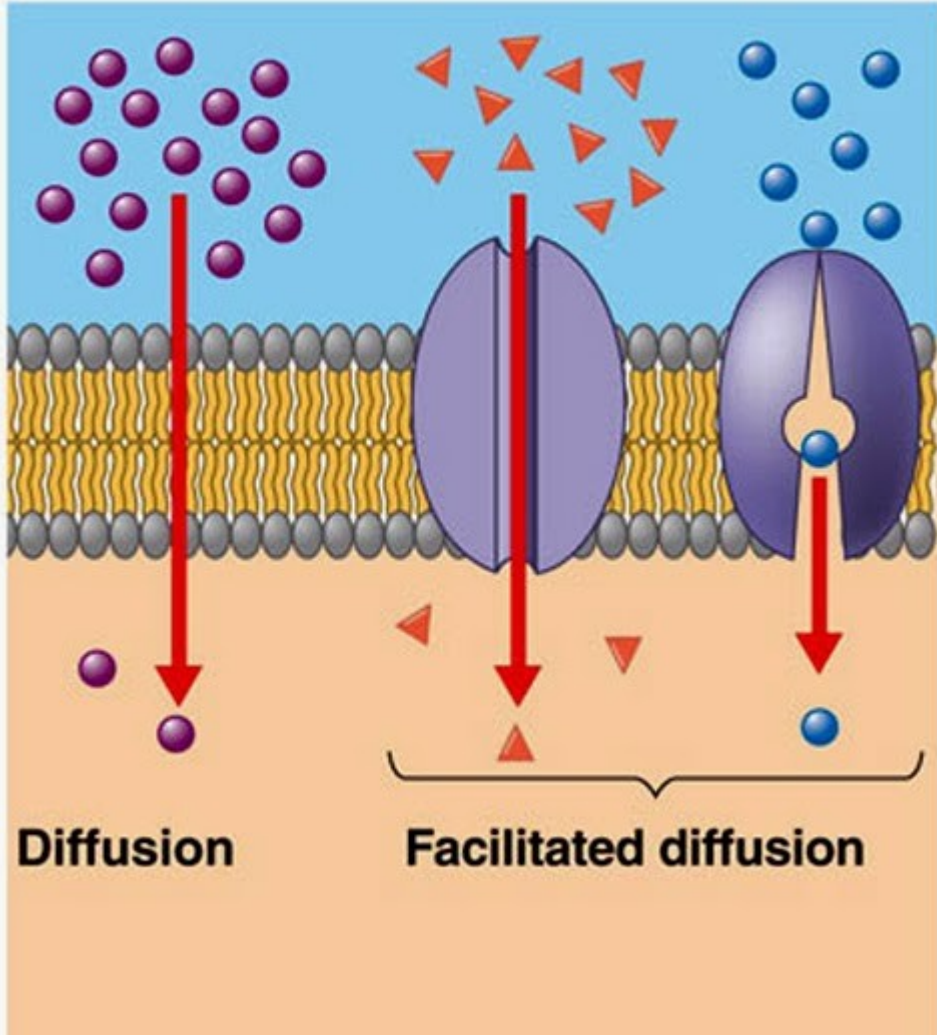
- **Anchoring:** cytoskeleton – supports structure/shape of cell.
- **Adhesion:** connects cells together, keeps cells where they belong.
- **Recognition:** HLA (Human Leukocyte **Antigens**) these are glycoproteins (partly sugar) that tell our immune system which cells are MINE – think NAME TAG!
- **Enzymes** – help chemical reactions happen.
- **Receptors** – respond to ligands (signaling molecules) that bind to them and cause something else to happen.
- **Carriers:** Opens and closes based on ligands binding to receptors.
- **Transmembrane** – moves 'stuff' (sugar, ions....) into and out of cell.
- **Channels:** (pores) always open – allows water to move via osmosis.

transport

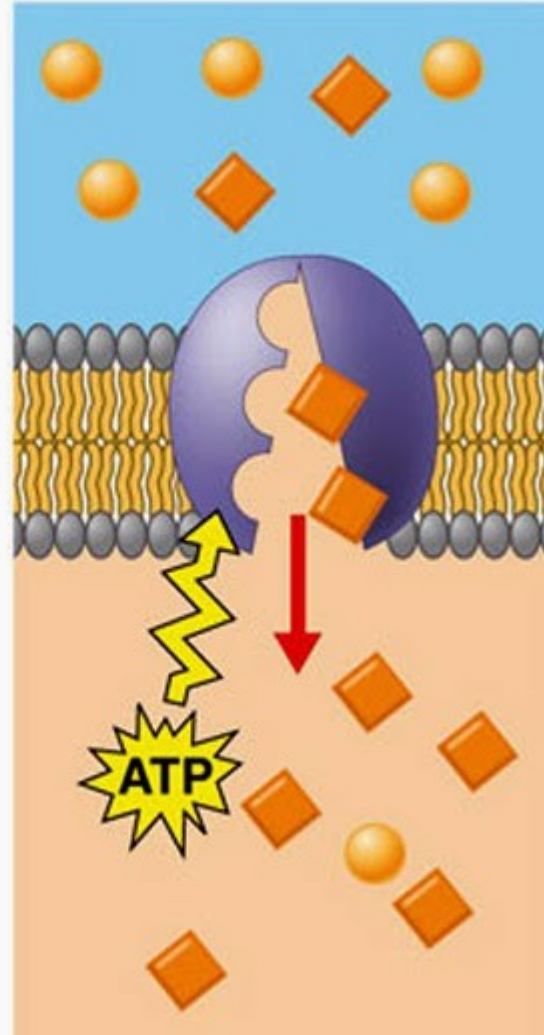
Phospholipid bilayer with proteins=cell/plasma membrane.
Main job of membrane = **Separate outside of cell from inside of cell**



Passive transport



Active transport



Diffusion & Active Transport

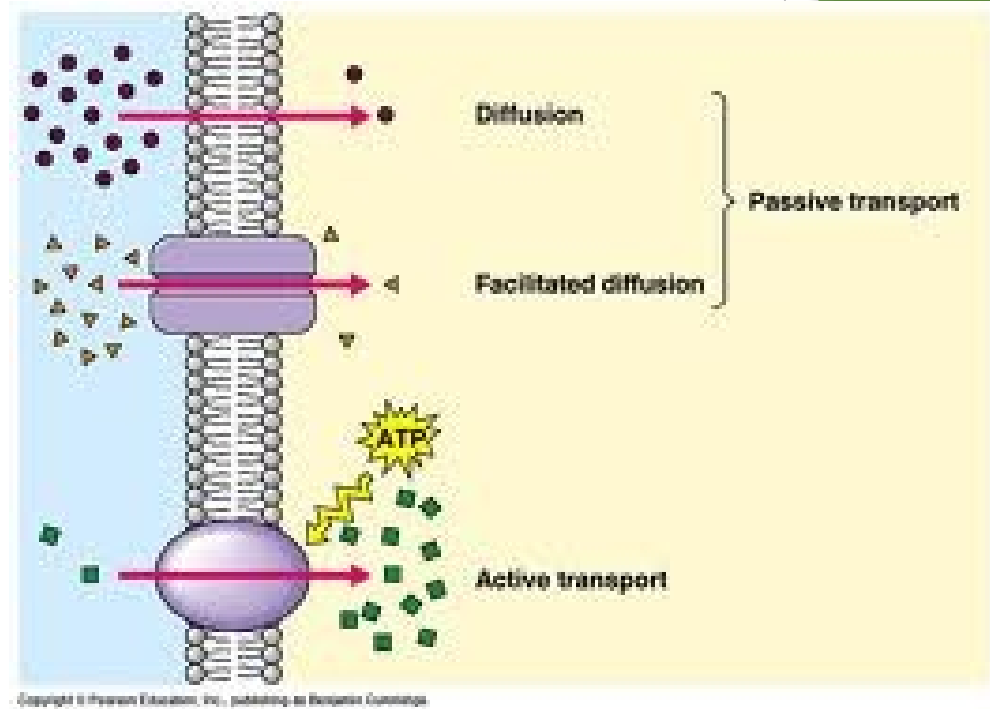
Diffusion does NOT use energy (ATP) – it always moves **from high** solute numbers **to low**.

Active needs energy (ATP) and moves **from low** solute numbers **to high**.

Diffusion

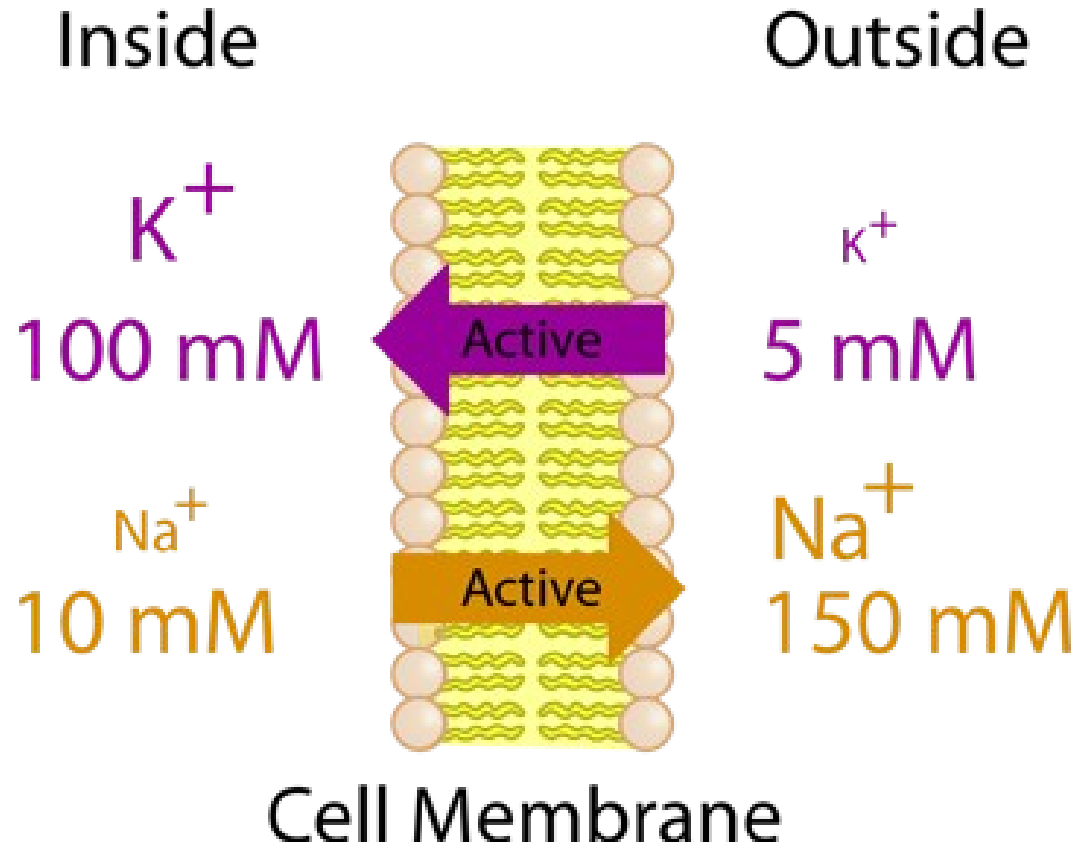
Molecules want to move DOWN/WITH the concentration gradient.

- SIMPLE diffusion occurs with gases – they move across the membrane.
- ACTIVE – notice the solutes are moving from few to high numbers – against the gradient



- Water moves via Osmosis – tries to equalize the concentration of solutes on each side. Most cells have “leak” channels for water.

Active Transport: (PUMP)

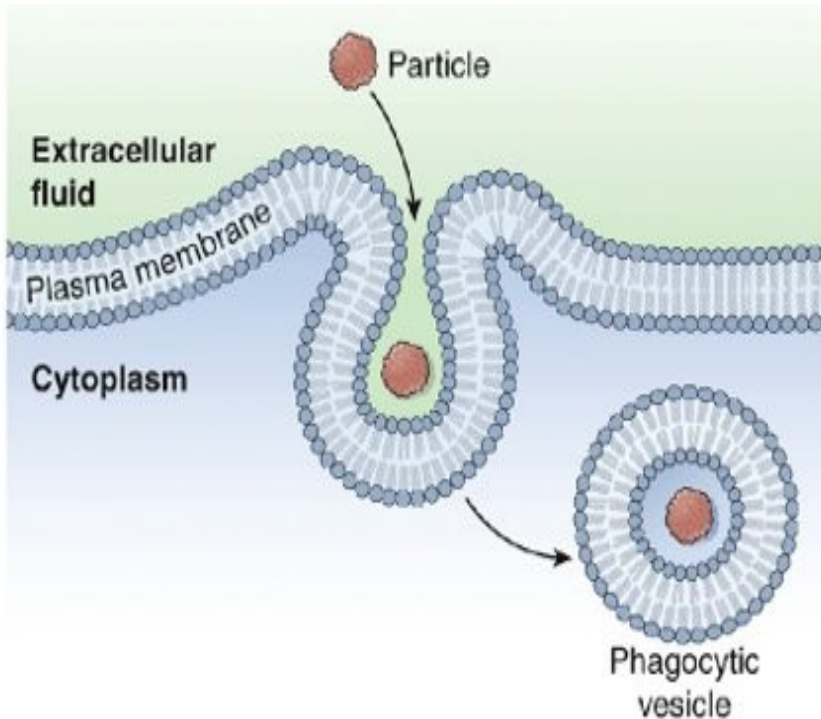


Notice that the molecules are moving from few to more, so energy (ATP) is needed! Against the concentration gradient.

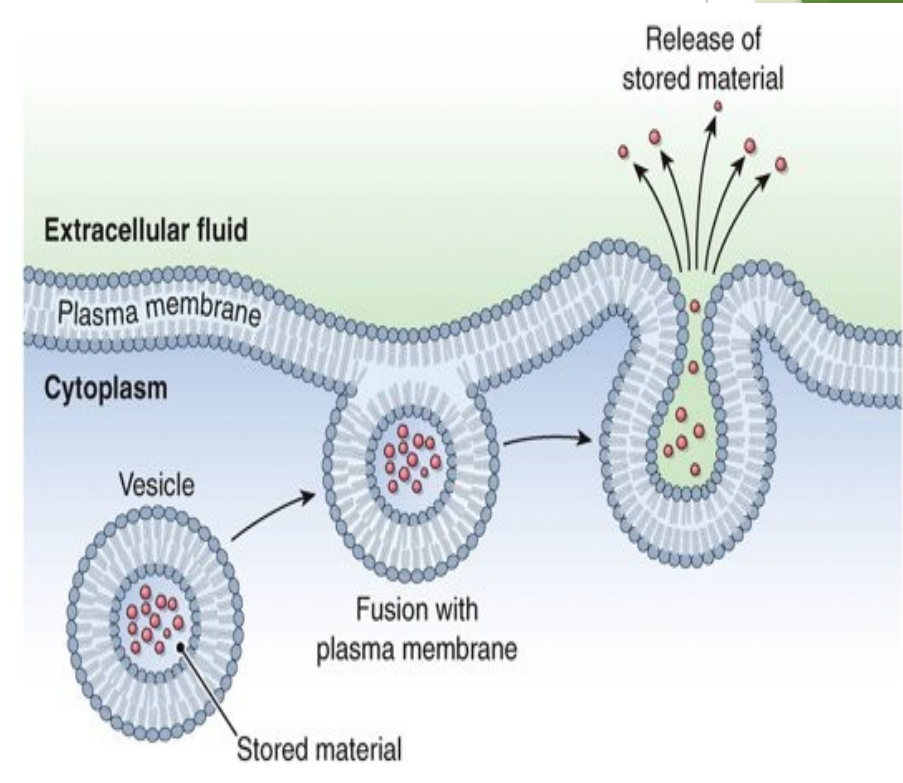
Sodium moves out where there is lots of it and potassium moves in where there is lots of it.

Vesicular Transport – A type of ACTIVE transport

Endocytosis – ENTERS
the cell



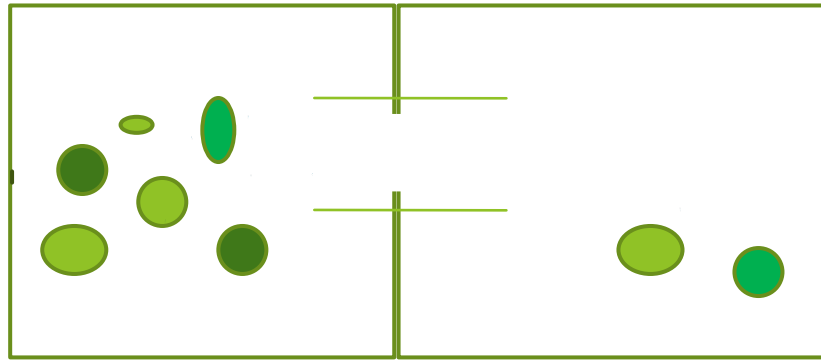
Exocytosis – EXITS the
cell



Osmosis

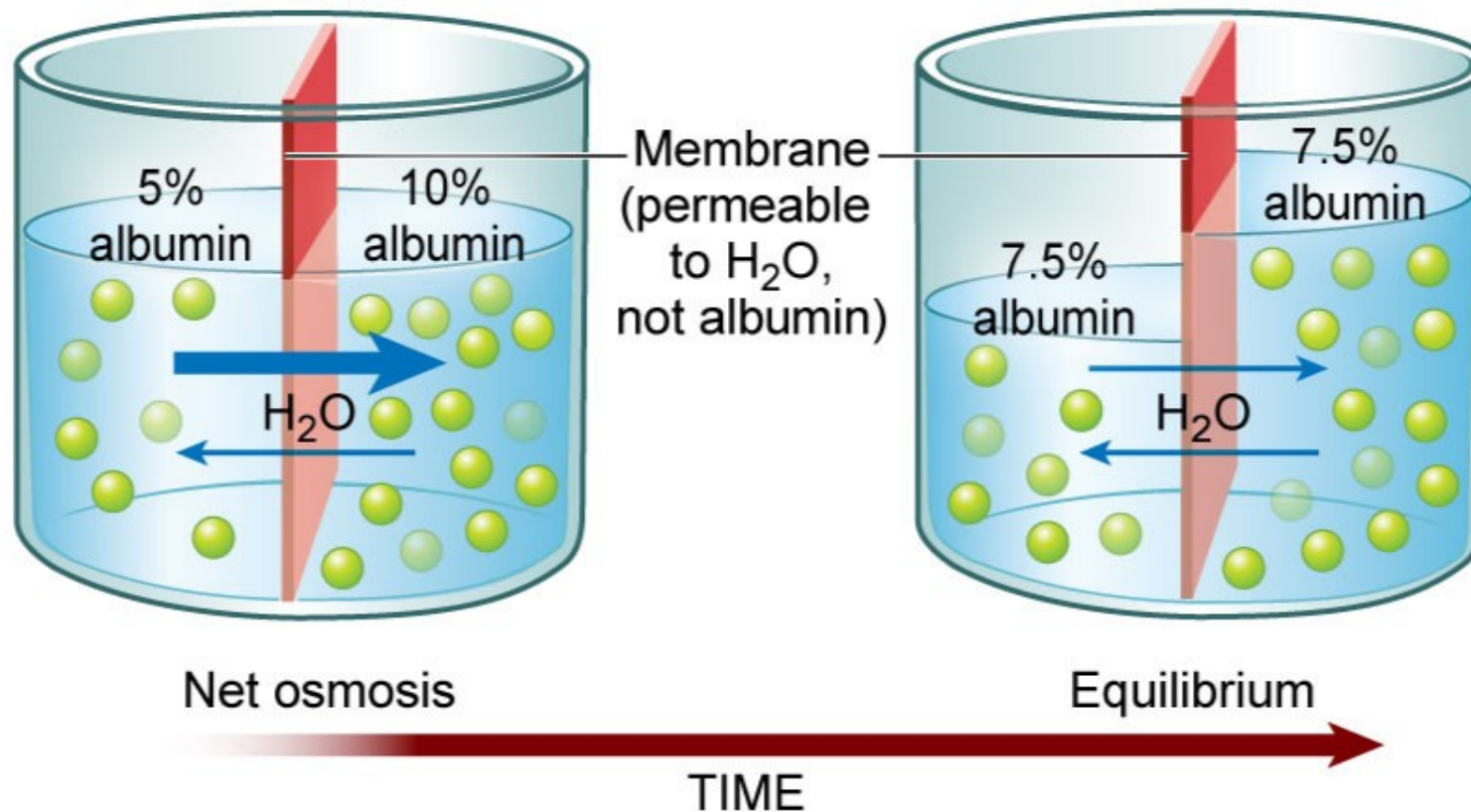
ONLY water moves!

Which way does the water move???? WHY?

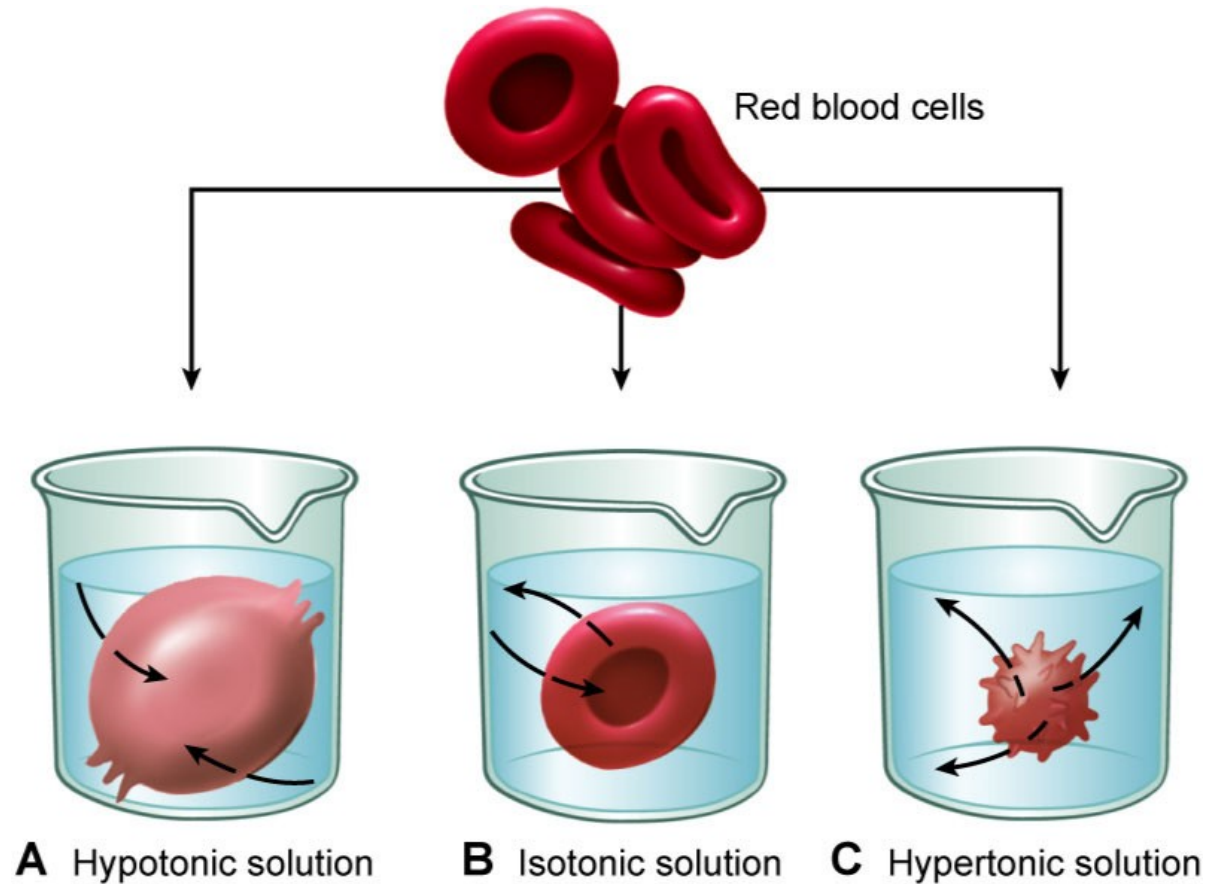


Water has a goal: **equalize** the **concentration** in the two compartments cells/ areas. **ONLY** water moves – not the solutes (molecules). The Channel/pore won't let the solutes move.

**Osmosis: concentrations need to be equal -
water will go toward more solutes!**



Why we care:



From Patton KT, Thibodeau GA: *Anatomy and physiology*, ed 7, St Louis, 2010, Mosby.

- Isotonic – same osmolality as ICF
- Hypotonic – lower osmolality – fewer solutes
- Hypertonic – higher osmolality – more solutes

Cell contains Organelles – they each have a specific job/function

Non – Membranous Organelles (Matching)

- ▶ 1- The **internal framework** of the cell.
 - ▶ 2- Organelle that makes/synthesizes proteins
 - ▶ 3- Greatly **increase the surface area** of membrane allowing for **greater absorption**.
 - ▶ 4- **Beat in unison** creating a unidirectional current that **propels a substance** across cell surfaces.
 - ▶ 5- **Form and organize spindle fibers during cell division.**
- ▶ A. Cilia
 - ▶ B. Microvilli
 - ▶ C. Cytoskeleton
 - ▶ D. Ribosome
 - ▶ E. Centrioles

Membranous Organelles:

Membrane Flow: all membranous are interconnected to communicate with each other.

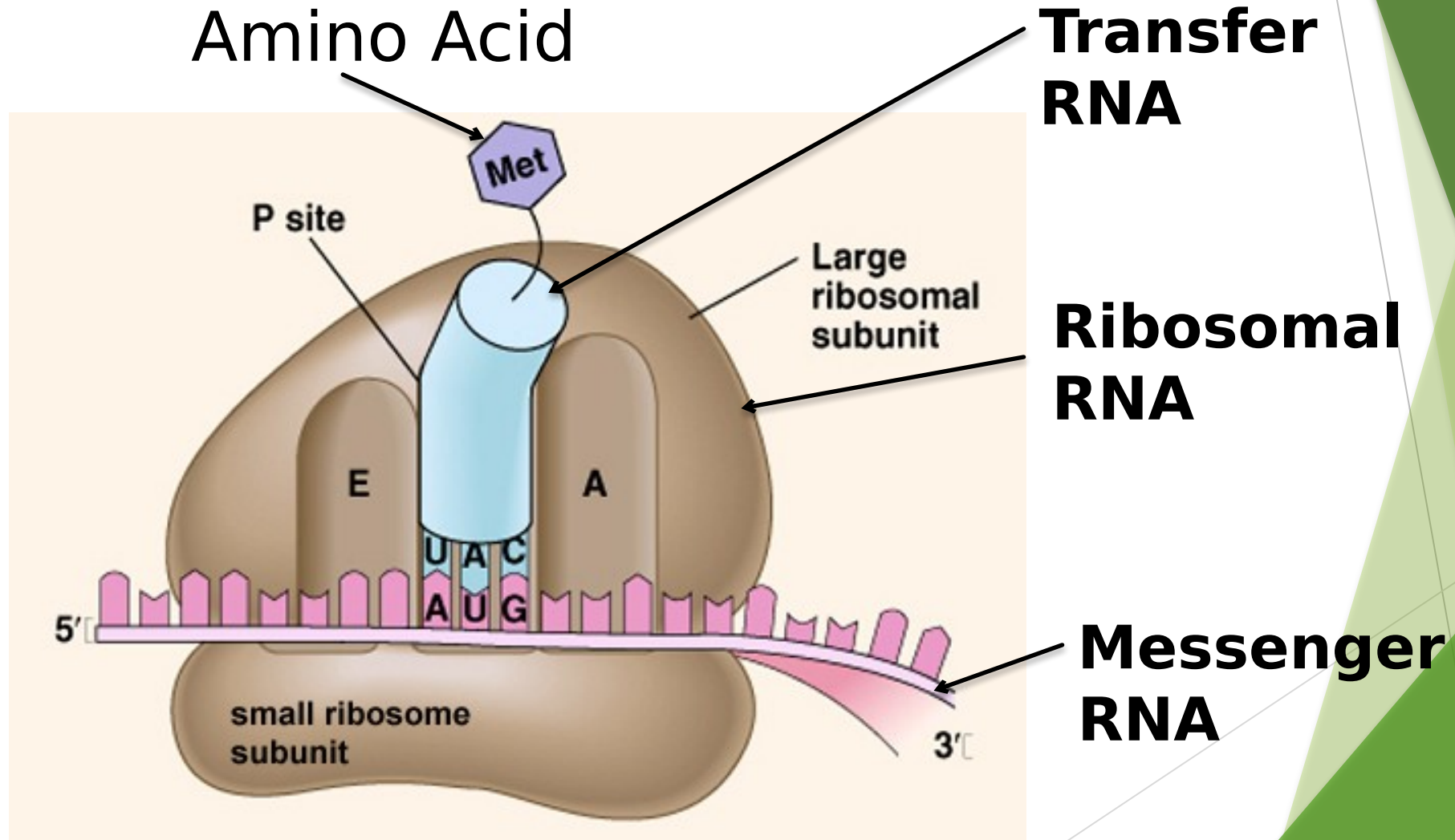
- Endoplasmic reticulum:
 - Rough - ribosomes (protein synthesis)
 - Smooth - Lipid synthesis
- Golgi - shipping center
- Lysosomes - Maid - DIGESTIVE enzymes
 - Peroxisomes - also maid - chemicals - hydrogen peroxide
- Mitochondria - ATP/energy production
- Nucleus - DNA - instructions for ALL cellular activity!

Nucleus, Ribosomes, ER, Golgi Making (Synthesizing) Proteins!!!

- Nucleus contains the information – the instructions on how to make the protein. – DNA
- This information is copied (**transcription**) into mRNA (messenger)
- mRNA moves out into the rough ER and finds a ribosome (assembly line)
- rRNA (ribosome) then works with tRNA to make a polypeptide (single strand of AA's) – this is called **translation**.
 - rRNA is the ribosome and it 'reads' the instructions carried by mRNA
 - tRNA brings in the parts (building blocks – AA) that matches the instructions.

A PRIMARY structure protein is made!

RNA - Protein synthesis

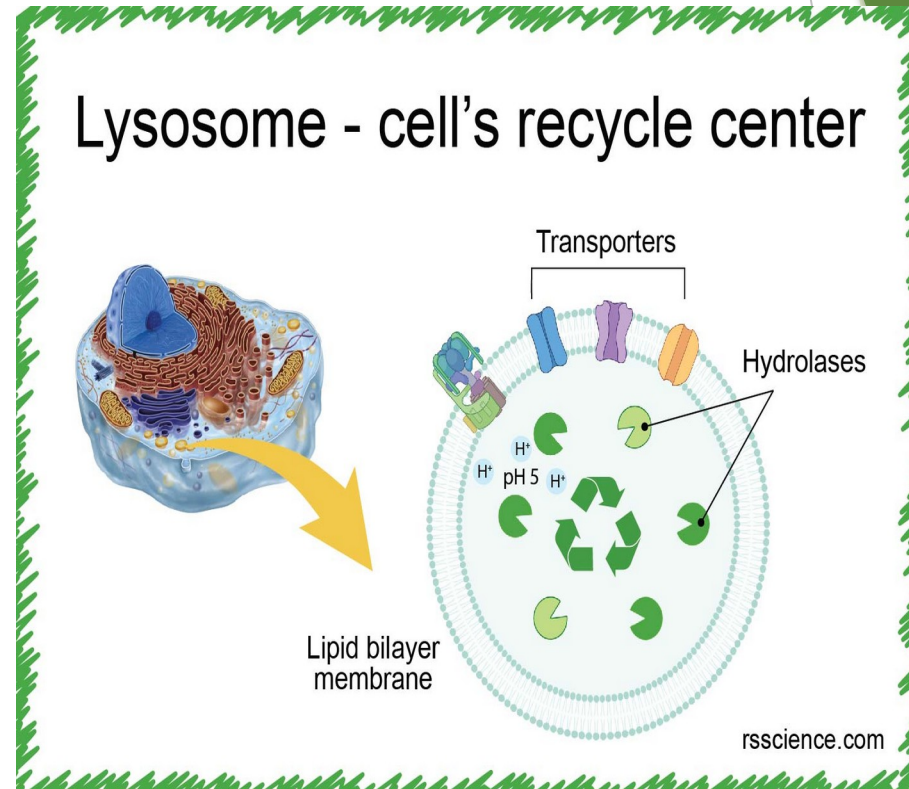


ER, Golgi - shipping Proteins!!!

- The primary protein then detaches from the ribosome & moves into the tube of the **Endoplasmic Reticulum** Cisternae (folds/tube)
 - In this area of the 'factory' this protein becomes secondary and tertiary (complex and refined)
 - Some have additional things added to them as per the instructions. (**Smooth ER**) **adds fats and MAKES fats - like phospholipids, triglycerides
- The prepared proteins (product made by this factory) gets an 'address' from the **Golgi** so it can go where it needs to go. (vesicles - 'boxes')
 - Some of them get shipped outside the cell - exocytosis
 - Some are used to repair the cells membrane
 - Some are used to make **lysosomes** - (maid)
 - Some become enzymes that the cell uses for its 'job'

Lysosome jobs: Full of **DIGESTIVE** enzymes

- 1- Keeps the inside of the cell clean by 'eating' damaged parts.
- 2- Eats foreign stuff from outside the cell. The cell engulfs this 'stuff' and the lysosome fuses and eats it.
Reuses the good/rejects the bad.
- 3- When it's time for the cell to die – the DNA initiates the autodestruct sequence and the lysosomes burst – releasing the digestive enzymes inside the cell – Autolysis/Apoptosis

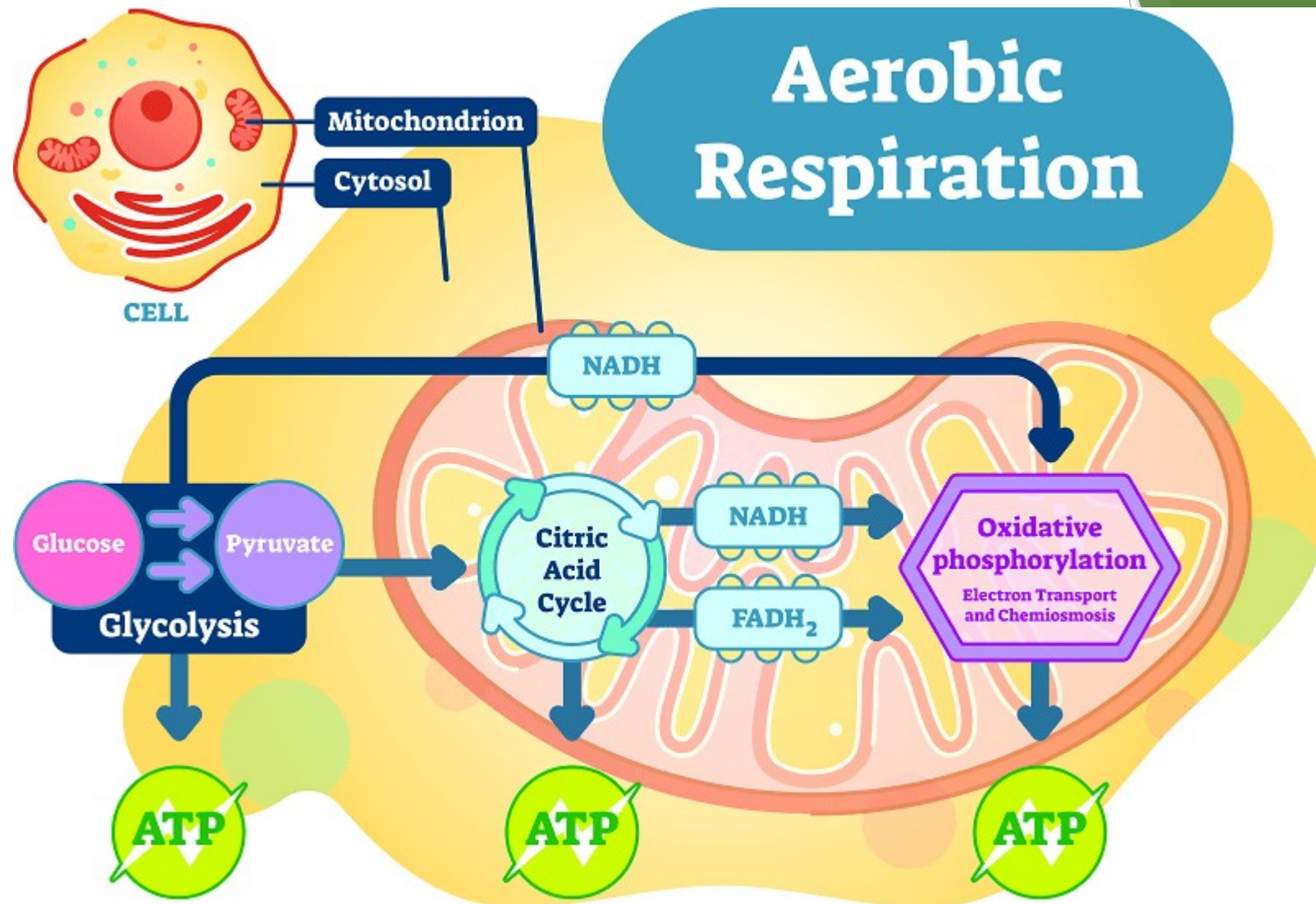


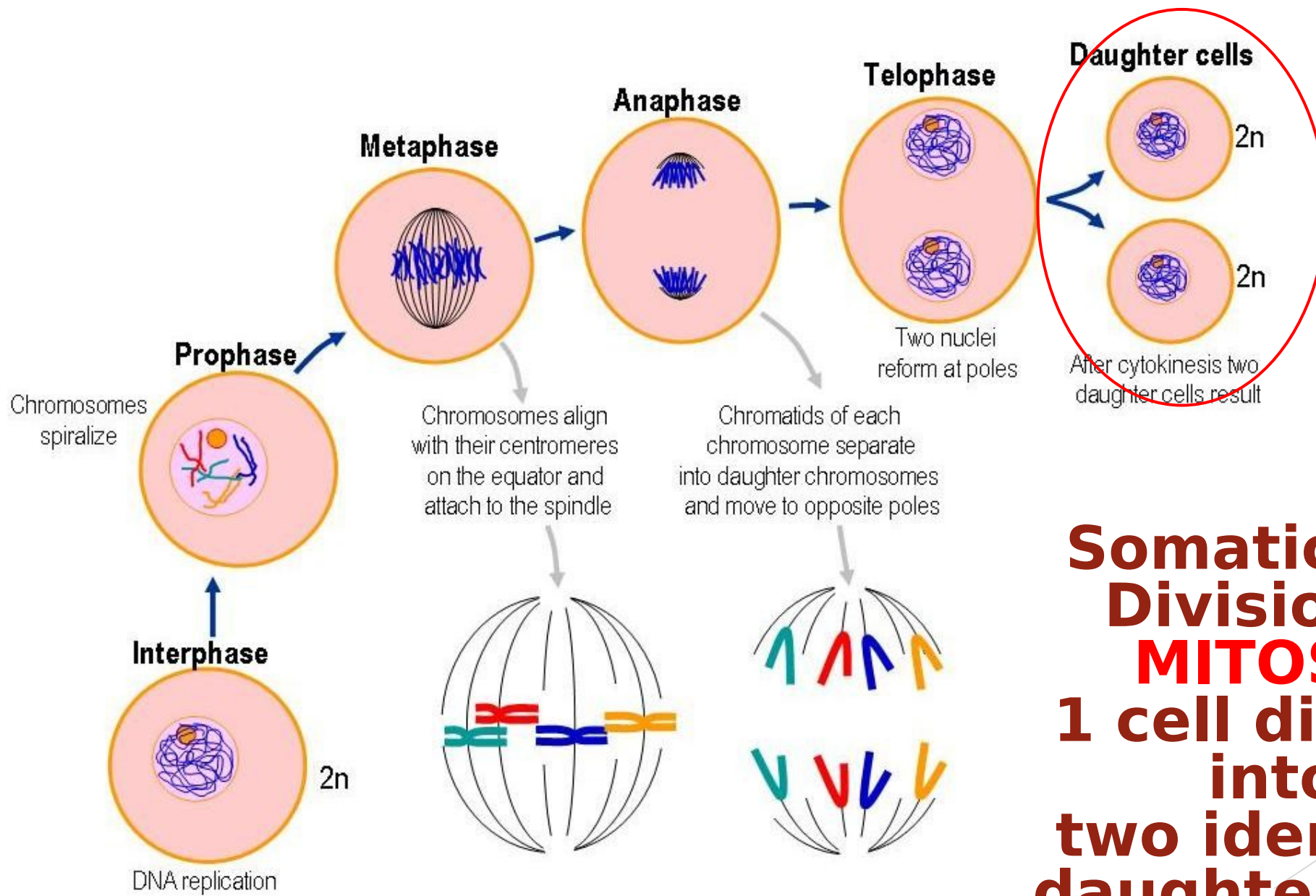
Mitochondria- Making ATP (“battery” full of energy)

- **Glycolysis**: one glucose breaks into two pyruvate (NO O₂ needed- anaerobic)
 - Happens in the **cytoplasm** – no organelle needed
- Pyruvate moves into the mitochondria (MUST have O₂ – aerobic)
- CO₂ is expelled which refines the pyruvate (makes it reactive)
- This moves into the **Krebs/Citric acid cycle** (a series of chemical reactions – inside the mitochondria matrix)
- H⁺ and O₂ combine to make water ** Oxygen is needed**
- AND the transfer of H⁺ phosphorylates ADP to make ATP – **Electron Transport chain** in the membrane of the mitochondria.

**Know these
steps**

Anaero
bic





**Somatic Cell
Division =
MITOSIS**
**1 cell divides
into
two identical
daughter cells**

Reminders:

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 2 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)
- Exam 1 opens Monday – covers modules 1 & 2

You can avoid feeling overwhelmed on Sunday by completing something each day - this also leads to better retention of knowledge.